



Louey Barbirou

Third-year Bachelor student in AI & Data · Data / AI apprenticeship · Available immediately
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Profile

Third-year Artificial Intelligence & Data student at Ynov Aix-en-Provence, available immediately for a data / AI apprenticeship (2 weeks in the company / 1 week at school). At trITon, I develop VideoInsight (AI-powered video analysis) and work on data preparation and analysis. I follow a product from the data all the way to deployment, and I don't stop at code that works: I also think about scalability, maintainability, performance and security. Versatile across data, development, infrastructure and security, and comfortable with development tools and AI assistants (I learned prompt engineering by building my own products), I want to join a team to prove what I can do and keep growing.

Technical skills

Programming languages : Python, R, SQL, SAS, C#, Java, PHP, JavaScript, Go

Data & Machine Learning : Pandas, NumPy, scikit-learn, XGBoost, Random Forest, regression, predictive modelling, statistics, forecasting, data visualization

AI & NLP : LLMs and multimodal AI (Claude, Gemini), natural language processing, text classification, applied prompt engineering (learned by building Prompt Booster)

Databases : SQL, NoSQL, MongoDB, PostgreSQL, Supabase

Business Intelligence : ETL, reporting, KPIs, dashboards, Apache Superset

DevOps & Tools : Docker, Git, Linux, Windows Server, FastAPI, Agile methodology

Security & infrastructure : Cybersecurity, bug bounty / web reconnaissance, phishing detection, Linux VPS deployment, containerisation, CI/CD (GitHub Actions, GitLab CI)

Dev & AI tools : VS Code, JetBrains, Git / GitHub, GitLab, Docker; Claude Code, GitHub Copilot, Windsurf, Gemini, Ollama (local LLMs), Hugging Face

Product mindset : Product vision, scalability, code maintainability, performance optimisation, shipping to production and monitoring

Soft skills & languages

Soft skills : Problem solving, project management, leadership, teamwork, communication

Languages : French (native), Arabic (native), English (fluent, near-native: spoken, written and read; preparatory programme taught in English)

Professional experience

AI & Data Development Intern

2026 — present

trITon — Aix-en-Provence, France (4-month internship)

- Development of VideoInsight, a web application for AI-powered video analysis and translation: transcription (Whisper), translation (MarianMT), dubbing / voice cloning (XTTS), lip synchronisation and automatic scene splitting.
- Building data preparation pipelines: cleaning and structuring the audio/video datasets used to train and evaluate the models.

- Implementation of a dashboard tracking quality indicators (transcription/translation accuracy, processing time) to monitor product performance.

Stock Manager & Flow Analyst

2023

Novogel, Sfax (Tunisia)

- Analysis of sales history to forecast supply and replenishment needs.
- Monitoring of stock rotation KPIs to reduce food waste.
- Set-up of tracking tools for stock traceability and profitability.

Hospitality & catering experience

2021 - 2026

Paupiette Groupe & Junior's (Marseille), Italian Trattoria (Aix-en-Provence)

- Head chef / commis / pizza chef: menu creation, stock management and rush-hour service (up to 400 covers), teamwork.

Projects

Source code, GitHub repos and live demos for every project: see my portfolio above.

WattCast — Electricity demand forecasting in production — model forecasting French electricity demand for the next day, scored daily against RTE's official forecast: automated daily pipeline, dbt warehouse, leakage-free walk-forward backtest 2023-2026, drift monitoring and retraining. 1.90% error (MAPE) vs 1.97% for RTE.

Stack Python, LightGBM, dbt, DuckDB, GitHub Actions, FastAPI, MLflow, Evidently

SupplyPulse — Supply Chain Analytics — supply chain analysis on real data (180,000 DataCo orders): layered data warehouse (raw → staging → marts) with DuckDB, governance (data dictionary + automated quality tests in CI) and a decision-support dashboard. Key insights: 54.8% late deliveries, ~10.8% margin.

Stack Python, DuckDB, SQL, Streamlit, pandas

Prompt Booster — web application that restructures a raw prompt (role, context, output format, constraints) to improve AI answers; building it is how I learned prompt engineering, which I then reused in my other products (AI extraction of quiz questions in Bac Quiz). Rule-based Python engine, deployed online.

Stack Python, Streamlit [Live demo](#)

Swarm Prediction — crop yield prediction (regression) on real FAO & World Bank data, plus a population simulation engine (SIR epidemiological model) with animated visualization.

Stack Python, pandas, NumPy, scikit-learn (Ridge, Random Forest), XGBoost

Olist — E-commerce BI dashboard — analysis of the Olist e-commerce dataset and a BI dashboard (KPIs: repeat purchase rate, delivery times) designed following data visualization principles (Few / Tufte).

Stack Python, SQL, PostgreSQL, Apache Superset, Docker

Bac Quiz — exam revision application: automatic multiple-choice question generation from PDF exam papers using multimodal (vision) AI, schema validation and a real-time backend.

Stack Python, multimodal AI (vision), Pydantic, Supabase, Expo / React Native

DBug Hunter — bug bounty assistance tool (web reconnaissance and security), deployed online.

Stack Python, Docker, Hugging Face Spaces [Live demo](#)

Phishing detection (email classification) — classification models to automatically detect malicious emails, as part of a project based on an industrial problem statement (Thales).

Stack Python, scikit-learn, NLP

Insight Hub — Student depression prediction — analysis and prediction of student depression on a 100,000-entry dataset: exploration of contributing factors (sleep, GPA, workload), interactive dashboard and a logistic regression prediction model.

Stack Python, scikit-learn, pandas, Plotly

Heart attack risk prediction — machine learning model estimating cardiac risk from clinical and lifestyle data; complete pipeline (preprocessing, scaling, training) and ROC-AUC evaluation.

Stack Python, scikit-learn, pandas

Trade With AI — crypto market prediction — crypto price prediction using time series (Prophet): trading signals (buy / sell / neutral), technical indicators and candlestick visualization exposed through an API.

Stack Python, Prophet, FastAPI

Education

Bachelor 3 — Artificial Intelligence & Data

2025 - present

Ynov, Aix-en-Provence, France

- Maths & Data Science (statistics, algebra, R, SAS, predictive modelling), advanced Python, databases (SQL, NoSQL, MongoDB).
- DevOps & management (Agile methodology), AI projects: hackathons, 48h Challenge, Ydays.

Bachelor 2 — Computer Science

2024 - 2025

Ynov, Aix-en-Provence, France

- Data & BI (advanced SQL, ETL, Python for Data, business intelligence), infrastructure & security (Docker, Windows Server, networks), C#, PHP, Java.

Bachelor 1 — IT fundamentals

2023 - 2024

Ynov, Aix-en-Provence, France

- Development & algorithms (Java, JavaScript, Go, OOP), infrastructure & systems (Linux, networks, SQL, Git).

Preparatory programme for engineering studies

2022 - 2023

South Mediterranean University, Tunisia

- Preparatory cycle taught entirely in English (North-American model).

High school diploma (Baccalaureate) — Computer Science major, with honours

2021 - 2022

Lycée 9 Avril, Sfax (Tunisia)